A weight-aware factor-base size for the ic planner (ic-planner/2)

2026-10-21 · study research/ic_planner_weight_rule_20261021 · protocol frozen in 7cf06ed6e, amendment 1 in 66f1642d1 (after Phase A, before Phase B) · seven ecbench sessions, 1,152 records, all verified, audits OK (12 exact replays each) · reference rho.negation.

Question

The planner's prime factor base, N_0 = round(#E^(1/3)/2) abscissae, is the cheapest size in counted S, but an earlier study found a doubled base 2–9% cheaper once the unpriced native work (lookups, guard probes, Legendre symbols, square roots, row operations) is charged at 1 GAE per operation (AGENTS.md §5a). Can the planner take a declared weight w and size the base for it, on curves the rule was not fitted on?

Phase A: derivation

Base multipliers m ∈ {0.5, 0.71, 1, 1.41, 2, 2.83, 4} on the registered curves icv1-fp24-tm5099-a6f137c6, icv1-fp26-tm1775-7e8fb6df and icv1-fp28-t28201-65099091. G(m, w) is the geometric mean over the three curves of S_w(m·N_0)/S_w(N_0), from ecbench band; the frozen rule takes m*(w) = argmin G.

w	0.5	0.71	1	1.41	2	2.83	4	m*
0	1.513	1.126	1	1.119	1.601	2.760	5.208	**1**
0.03	1.536	1.138	1	1.103	1.558	2.668	5.019	**1**
0.1	1.584	1.161	1	1.072	1.471	2.478	4.629	**1**
0.3	1.686	1.210	1	1.005	1.287	2.076	3.801	**1**
1	1.858	1.295	1	**0.890**	0.972	1.392	2.394	**1.41**

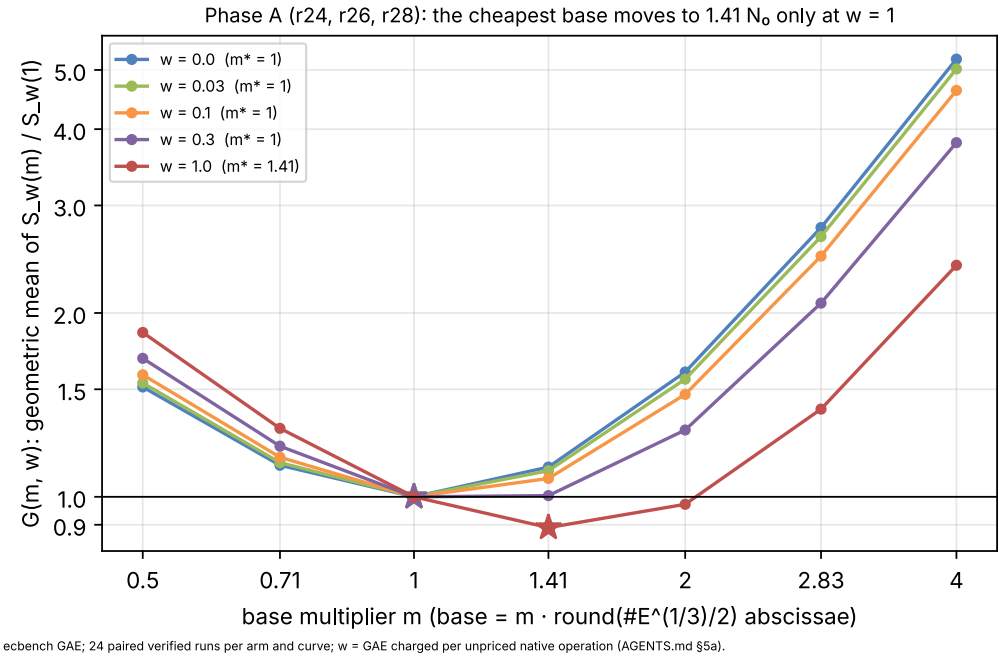


Figure 1: Phase A: G(m, w). The minimum moves off m = 1 only at w = 1.

Phase B: held out

Four registered prime-order curves in none of the planner's evidence; v1 = N_0 against v2-m141 = round(1.41·N_0). Paired S(v2-m141)/S(v1), ecbench compare --weight, cluster 95% intervals, 24 pairs each:

curve	$\log_2 r$	$N_0 \to$ v2	w = 0	w = 1
icv1-fp21-t377-d7bba3fb	20.59	58 → 82	1.040 [0.911, 1.176]	0.850 [0.713, 1.002]
icv1-fp22-tm2947-e15fed69	21.19	67 → 94	1.058 [0.956, 1.167]	0.855 [0.743, 0.972]
icv1-fp24-t2705-27cbb01b	23.38	111 → 157	0.996 [0.884, 1.113]	0.759 [0.642, 0.883]
icv1-fp29-t47-14392892	28.25	342 → 482	1.062 [0.993, 1.144]	0.819 [0.749, 0.900]

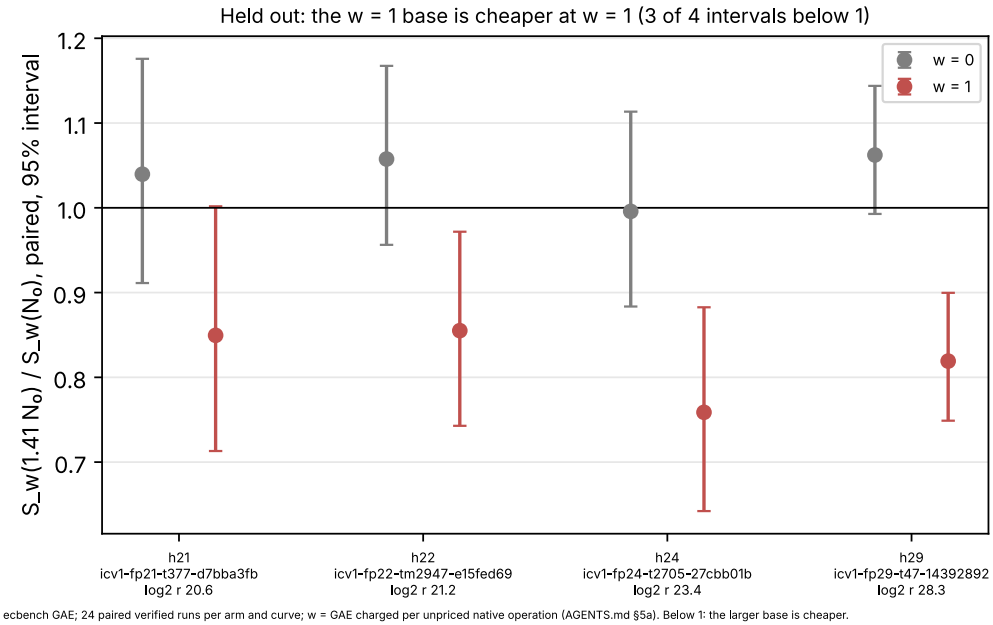


Figure 2: Held out: the w = 1 base against N_0 at w = 0 and w = 1.

Predictions. P1 (m* non-decreasing in w) held. P2 (m*(0) = 1, m*(1) ∈ {1.41, 2, 2.83}) held. P3 (held out, w = 1, interval below 1 on at least 3 of 4) held, 3 of 4; h21′s reaches 1.002. P4 (held out, w = 0, v2 ≥ v1) not strictly held: h24 reads 0.996 and every w = 0 interval contains 1. The larger base's counted price is within noise there, and 8.5–14.4% (every interval above 1) on the derivation curves.

Distance remaining to rho

`S(IC)/S(rho.negation)`, paired; rho pays its native work at w = 1 too. Rho is cheaper everywhere. On the held-out curves the plan sits 9.0–18.3× rho at w = 0. At w = 1 the `N₀` base reads 10.6–21.9× and the weighted base 9.1–17.9×: the weighted base closes part of the gap that pricing the native work opens. These are distances remaining, not a pass/fail gate.

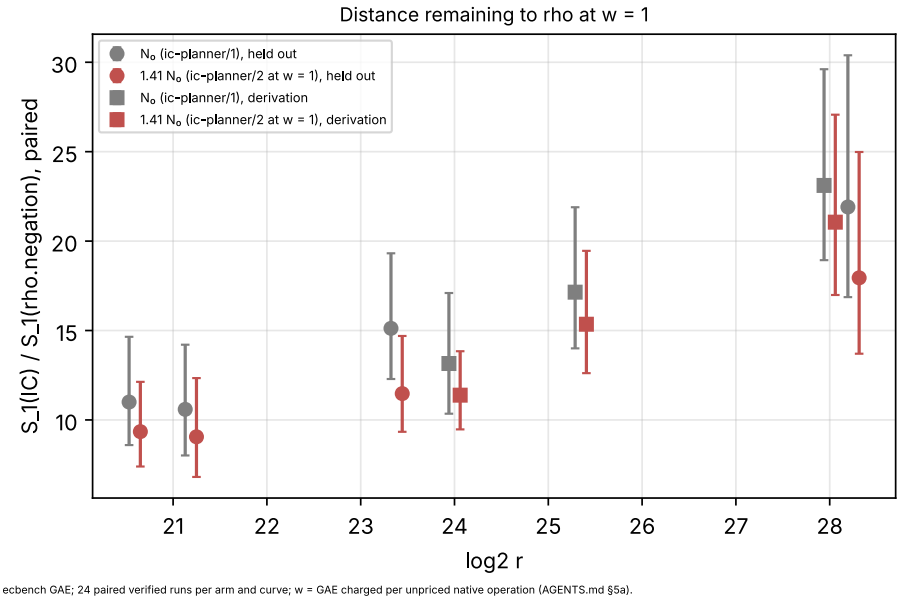


Figure 3: S_1(IC)/S_1(rho.negation) for both base sizes.

Decision and class

Adopted as `ic-planner/2` under the decision rule (P3): `ic plan --native-weight w` and `ic bench --auto --native-weight w` use `round(m*(w)·N₀)`, `m*` read at the largest measured weight not above w. The default (w = 0) is unchanged, as are every other stage and every non-prime class. Engineering (AGENTS.md §3): the weight is a declared pricing assumption, and a w = 1 plan is cheaper only at w = 1, never a ranking across the §5a band. No counted S improves on

a measured baseline, so no row of tables A–C moves. The validation gate needs an independent replay certificate and validator report, so no leaderboard or `benchmarks/README.md` row is added here.

Deviations: cards 2 and 3 stopped before writing any session file (ecbench refused a busy host, PSI 6.02 and 6.37 > 5); card 4 waits for 30 s of quiet; specs, seeds and binary unchanged. `1.41 × 128 = 180.48` was run as 180.